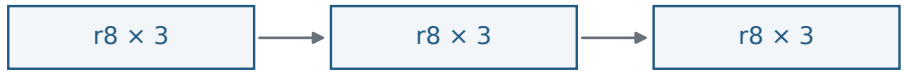
1 Workload and dependencies

One chain



Four chains

Each register feeds its next use; this four-instruction group repeats eight times.

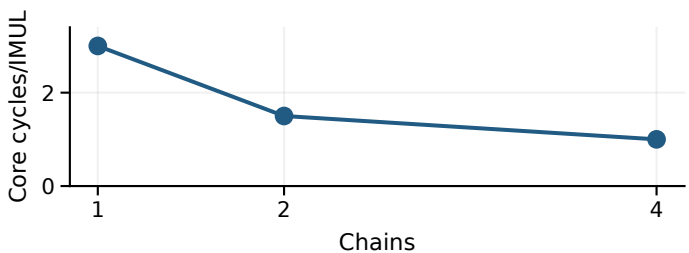
2 Controlled comparison

Same 32 IMULs/loop; change only 1, 2 or 4 destination chains.

3 Timed batch and normalization

Host timer around I=100,000 loops; ticks/IMUL = elapsed TSC / (32 I).

4 Measured response



5 Inference and control

A dependent result takes about three cycles, while independent work initiates about once per cycle. Fixed work and the separately cited single eligible IMUL resource model support pipelining; timing alone does not count replicated units.